

## Project Initialization and Planning Phase

|               |                                                            |
|---------------|------------------------------------------------------------|
| Date          | 04 July 2026                                               |
| Team ID       | xxxxxxx                                                    |
| Project Name  | Opticrop-Smart Agricultural Production Optimization Engine |
| Maximum Marks | 3 Marks                                                    |

### Define Problem Statements :

Farmers often face challenges in selecting the most suitable crop due to varying soil and environmental conditions. Factors such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH level, and rainfall significantly impact crop growth and productivity. Traditional farming decisions are often based on experience and manual assessment, which may lead to poor crop selection, low yields, inefficient resource utilization, and financial losses. Therefore, there is a need for an intelligent and data-driven system that can analyze these factors and recommend the most suitable crop. The Smart Agricultural Production Optimization Engine (OptiCrop) addresses this challenge by using machine learning techniques to provide accurate crop recommendations, improve productivity, optimize resource usage, and support sustainable agricultural practices.

Example:



| Problem Statement                                                          | I am (Customer)                                    | I'm trying to                                                   | But                                                                  | Because                                                      | Which makes me feel                            |
|----------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------|
| Farmers struggle to select the right crop and achieve higher productivity. | A farmer looking to improve crop yield and profit. | Find the most suitable crop for my soil and climate conditions. | I often choose the wrong crop and face low yield and financial loss. | I lack accurate data and expert guidance for crop selection. | Worried and unsure about my farming decisions. |